

CONGA Model

UML object model (io.wcm.devops.conga.model) — reproduced as selectable text; verified against branch develop / v1.20 (2026-08).

Role

- inherits : List<RoleInherit>

- variants : List<RoleVariant>

- templateDir : String

- files : List<RoleFile>

- sensitiveConfigParameters

(extends AbstractConfigurable)

RoleVariant

- variant : String

(+ config : Map)

RoleFile

extends AbstractModel (not Configurable)

- file : String

- dir : String

- template : String (Handlebars .hbs)

- url : String

- symlinkTarget : String

- variants : List<String>

- condition : String

- validators : List<String>

- validatorOptions : Map

- postProcessors : List<String>

- postProcessorOptions : Map

- fileHeader : String

- multiply : String

- multiplyOptions : Map

- charset : String = UTF-8

- lineEndings : LineEndings = unix

- escapingStrategy : String

- modelOptions : Map

- deleteSource : boolean

Environment

- nodes : List<Node>

- roleConfig : List<RoleConfig>

- tenants : List<Tenant>

- dependencies : List<String>

- pluginConfig : Map

(extends AbstractConfigurable)

RoleConfig

- role : String

(+ config : Map)

NodeRole

- role : String

- variant / variants

(+ config : Map)

Node

- node : String

- nodes : List<String>

- roles : List<NodeRole>

(+ config : Map)

Tenant

- tenant : String

- roles : List<String>

tenant roles (≠ node roles)

RoleInherit

- role : String

(target role to inherit)

AbstractConfigurable

- config : Map<String,Object> = new HashMap<>()

merged along the tree — see merge order at the bottom ↓

«interface» Configurable

+ config : Map<String,Object>

ContextProperties (implicit — set by CONGA, not in YAML)

VERSION = "version"

ROLE = "nodeRole"

ROLE_VARIANT = "nodeRoleVariant"

ENVIRONMENT = "environment"

NODE = "node"

NODES = "nodes"

NODES_BY_ROLE = "nodesByRole"

NODES_BY_ROLE_VARIANT= "nodesByRoleVariant"

TENANTS = "tenants"

TENANTS_BY_ROLE = "tenantsByRole"

TENANT = "tenant"

TENANT_ROLES = "tenantRoles"

OPS — Environment (two apaches)

two apache nodes, each 1 role

nodes:

- nodes:

- myApache

config:

varX: v4_NodeCfg

roles:

- role: webserver

variant: stdVariant

config:

varX: v6_NodeRoleCfg

- nodes:

- mySslApache

config:

varX: v4_NodeCfg-ssl

roles:

- role: webserver

variant: secureVariant

config:

varX: v6_NodeRoleCfg-ssl

role-specific global config

roleConfig:

- role: webserver

config:

varX: v5_RoleCfg

global configuration

config:

varX: v3_EnvironmentCfg

tenant definitions

tenants:

- tenant: tenant1

roles:

- webserver

config:

varX: v7_Tenant1Cfg-de

website.domain: mysite.de

- tenant: tenant2

roles:

- webserver

config:

varX: v7_Tenant2Cfg-fr

website:

domain: mysite.fr

DEV — defines a role

variants:

- variant: stdVariant

config:

varX: v2_RoleVariantStd

- variant: secureVariant

config:

varX: v2_RoleVariantSsl

templateDir: webserver

files:

- file: "\${tenant}_vhost.conf"

dir: vhosts

template: tenant_vhost.conf.hbs

variants:

- stdVariant

multiply each tenant w/ role

multiply: tenant

multiplyOptions:

roles:

- webserver

- file: "\${tenant}_vhost_ssl.conf"

dir: vhosts

template: tenant_vhost_ssl.conf.hbs

variants:

- secureVariant

multiply: tenant

multiplyOptions:

roles:

- webserver

role parameters and defaults

config:

varX: v1_RoleCfg-web

eg.

{{#each tenantsByRole.webserver}}{{this.config.website.domain}}{{#unless @last}},{{/unless}}{{/each}}

→ becomes "mysite.de,mysite.fr"

